ADVANCED
Stage 6

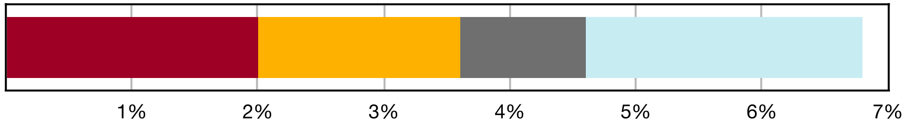
Trigonometry (Adv), T1 Trigonometry and Measure of Angles (Adv)

Bearings (Y11)

Teacher: SHS Maths

Exam Equivalent Time: 70.5 minutes *(based on allocation of 1.5 minutes per mark)*

T1 Trigonometry and Measure of Angles



- Trig Ratios, Sine and Cosine Rules
- 3D Trigonometry
- Bearings
- Circular Measure

***SmarterMaths analytics based on the average contribution to new syllabus Advanced Maths exams since 2020.**

HISTORICAL CONTRIBUTION

- *T1 Trigonometry and Measure of Angles* has contributed a healthy average of 6.8% per Advanced exam since the new syllabus was introduced in 2020.
- This topic has been split into four sub-topics for analysis purposes: *1-Trig Ratios, Sine and Cosine Rules (2.0%), 2- 3D Trigonometry (1.6%), 3-Bearings (1.0%) and 4-Circular Measure (2.2%).*
- This analysis looks at *Bearings.*

HSC ANALYSIS - What to expect and common pitfalls

- *Bearings (1.0%)* has only been examined once in new syllabus *Advanced* exams to date, although the question was worth a substantial 5 marks *(2020 Adv 15).*
- *Bearings* is a prime topic area to test Advanced and Standard 2 common content and is due a meaningful mark allocation, in our view.
- Before 2020, bearings was most recently tested in the *Advanced exam* in 2018 and 2014, and well answered.
- A number of Std2 past HSC questions that we consider good revision have been included in this database (identified by 2ADV* or STD2 in their title).

Questions

1. Measurement, STD2 M6 SM-Bank 2 MC

Which of the following expresses S25°E as a true bearing?

A. 025°

B. 065°

C. 115°

D. 155°

2. Measurement, STD2 M6 SM-Bank 3 MC

Which of the following expresses S65°W as a true bearing?

A. 065°

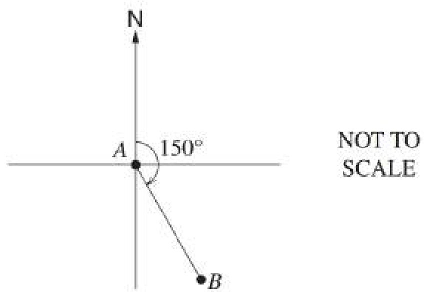
B. 155°

C. 245°

D. 295°

3. Trigonometry, 2ADV* T1 2010 HSC 10 MC

A plane flies on a bearing of 150° from A to B.



What is the bearing of A from B?

A. 30°

B. 150°

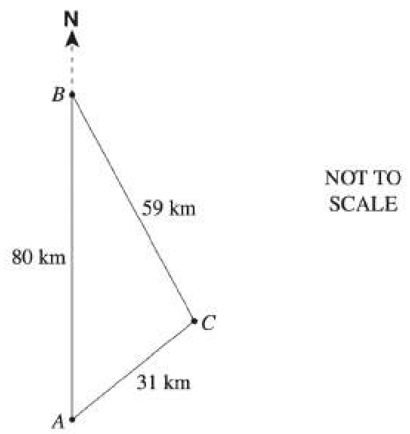
C. 210°

D. 330°

4. Trigonometry, 2ADV* T1 2012 HSC 20 MC

Town B is 80 km due north of Town A and 59 km from Town C.

Town A is 31 km from Town C.



What is the bearing of Town C from Town B?

A. 019°

B. 122°

C. 161°

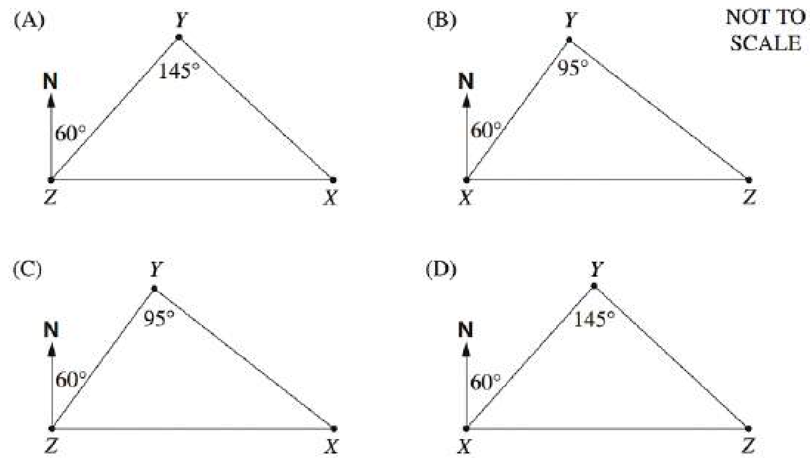
D. 341°

5. Trigonometry, 2ADV* T1 2014 HSC 23 MC

The following information is given about the locations of three towns X, Y and Z:

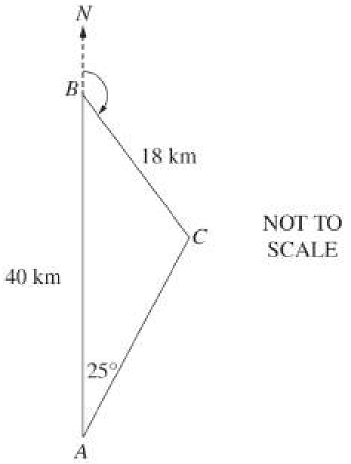
- X is due east of Z
- X is on a bearing of 145° from Y
- Y is on a bearing of 060° from Z.

Which diagram best represents this information?



6. Trigonometry, 2ADV* T1 2016 HSC 25 MC

The diagram shows towns A, B and C. Town B is 40 km due north of town A. The distance from B to C is 18 km and the bearing of C from A is 025°. It is known that $\angle BCA$ is obtuse.



What is the bearing of C from B?

A. 070°

B. 095°

C. 110°

D. 135°

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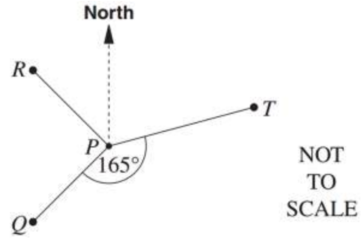
7. Trigonometry, 2ADV* T1 2008 HSC 17 MC

The diagram shows the position of Q, R and T relative to P.



In the diagram,

Q is south-west of P

R is north-west of P

$\angle QPT$ is 165°

What is the bearing of T from P?

A. 060°

B. 075°

C. 105°

D. 120°

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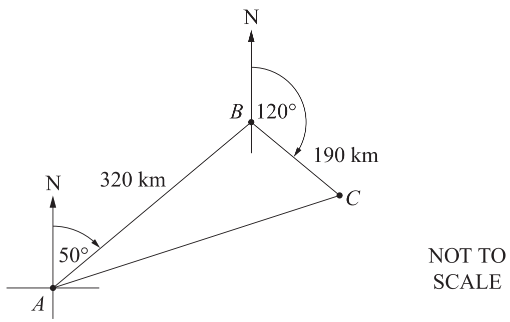
8. Trigonometry, 2ADV T1 2018 HSC 12a

A ship travels from Port *A* on a bearing of 050° for 320 km to Port *B*. It then travels on a bearing of 120° for 190 km to Port *C*.

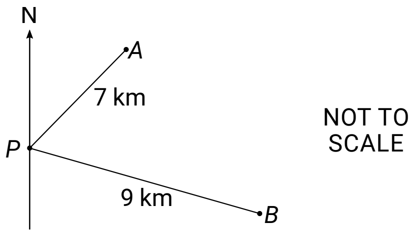


i. What is the size of $\angle ABC$? ***(1 mark)***

ii. What is the distance from Port *A* to Port *C* ? Answer to the nearest 10 kilometres. ***(2 marks)***

9. Trigonometry, 2ADV T1 2020 HSC 15

Mr Ali, Ms Brown and a group of students were camping at the site located at P. Mr Ali walked with some of the students on a bearing of 035° for 7 km to location A. Ms Brown, with the rest of the students, walked on a bearing of 100° for 9 km to location B.



a. Show that the angle APB is 65°. ***(1 mark)***

b. Find the distance AB. ***(2 marks)***

c. Find the bearing of Ms Brown's group from Mr Ali's group. Give your answer correct to the nearest degree. ***(2 marks)***

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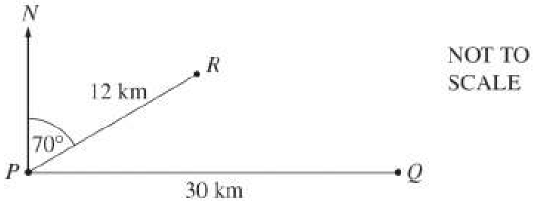
10. Trigonometry, 2ADV T1 2004 HSC 3c



The diagram shows a point P which is 30 km due west of the point Q.

The point R is 12 km from P and has a bearing from P of 070°.

i. Find the distance of R from Q. ***(2 marks)***

ii. Find the bearing of R from Q. ***(2 marks)***

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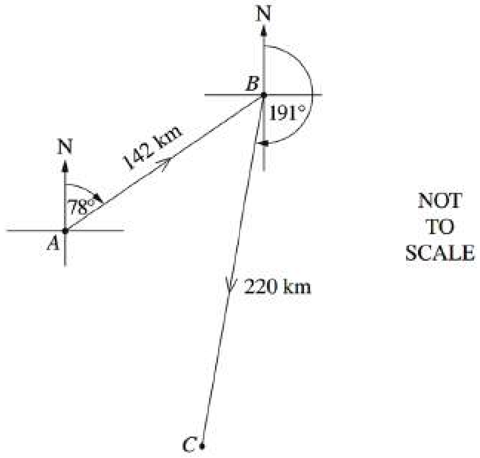
11. Trigonometry, 2ADV T1 2014 HSC 13d

Chris leaves island A in a boat and sails 142 km on a bearing of 078° to island B. Chris then sails on a bearing of 191° for 220 km to island C, as shown in the diagram.



i. Show that the distance from island C to island A is approximately 210 km. ***(2 marks)***

ii. Chris wants to sail from island C directly to island A. On what bearing should Chris sail? Give your answer correct to the nearest degree. ***(3 marks)***

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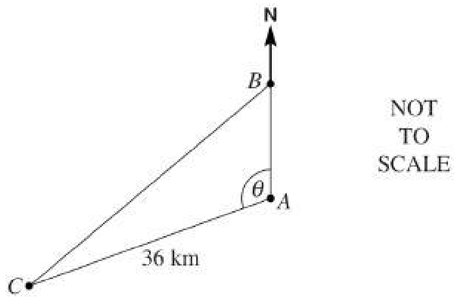
12. Trigonometry, 2ADV* T1 2005 HSC 27c



The bearing of C from A is 250° and the distance of C from A is 36 km.

i. Explain why θ is 110°. ***(1 mark)***

ii. If B is 15 km due north of A, calculate the distance of C from B, correct to the nearest kilometre. ***(3 marks)***

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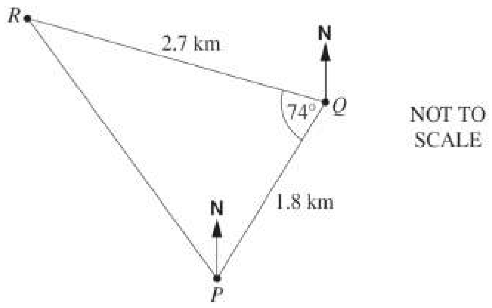
13. Trigonometry, 2ADV* T1 2009 HSC 27b

A yacht race follows the triangular course shown in the diagram. The course from P to Q is 1.8 km on a true bearing of 058°.

At Q the course changes direction. The course from Q to R is 2.7 km and $\angle PQR = 74°$.



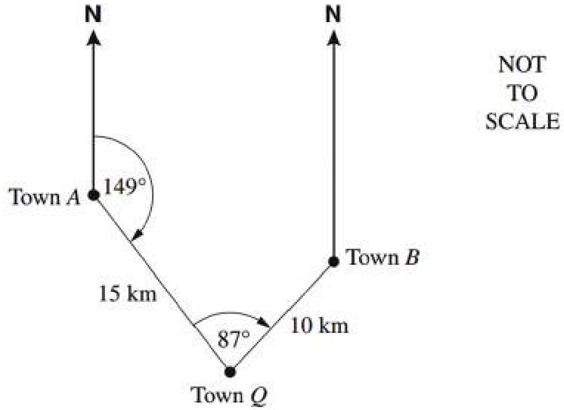
i. What is the bearing of R from Q? ***(1 mark)***

ii. What is the distance from R to P? ***(2 marks)***

iii. The area inside this triangular course is set as a 'no-go' zone for other boats while the race is on. What is the area of this 'no-go' zone? ***(1 mark)***

14. Trigonometry, 2ADV* T1 2007 26a

The diagram shows information about the locations of towns A, B and Q.



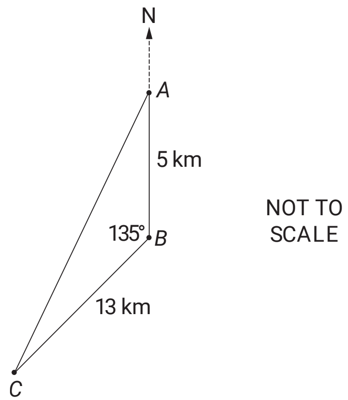
i. It takes Elina 2 hours and 48 minutes to walk directly from Town A to Town Q. Calculate her walking speed correct to the nearest km/h. ***(1 mark)***

ii. Elina decides, instead, to walk to Town B from Town A and then to Town Q. Find the distance from Town A to Town B. Give your answer to the nearest km. ***(2 marks)***

iii. Calculate the bearing of Town Q from Town B. ***(1 mark)***

15. Trigonometry, 2ADV* T1 2017 HSC 30c

The diagram shows the location of three schools. School A is 5 km due north of school B, school C is 13 km from school B and $\angle ABC$ is 135°.

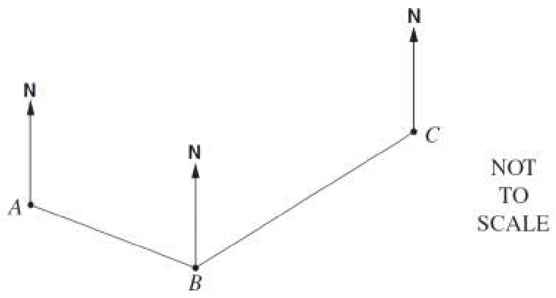


i. Calculate the shortest distance from school A to school C, to the nearest kilometre. ***(2 marks)***

ii. Determine the bearing of school C from school A, to the nearest degree. ***(3 marks)***

16. Trigonometry, 2ADV* T1 2011 HSC 24c

A ship sails 6 km from A to B on a bearing of 121°. It then sails 9 km to C. The size of angle ABC is 114°.



Copy the diagram into your writing booklet and show all the information on it.

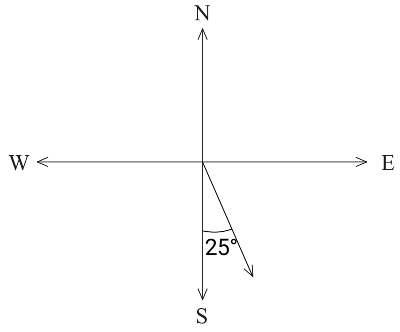
i. What is the bearing of C from B? ***(1 mark)***

ii. Find the distance AC. Give your answer correct to the nearest kilometre. ***(2 marks)***

iii. What is the bearing of A from C? Give your answer correct to the nearest degree. ***(3 marks)***

Worked Solutions

1. Measurement, STD2 M6 SM-Bank 2 MC



$$\begin{aligned} \text{S}60^{\circ}\text{W} &= 180 - 25 \\ &= 155^{\circ} \end{aligned}$$

$\Rightarrow D$

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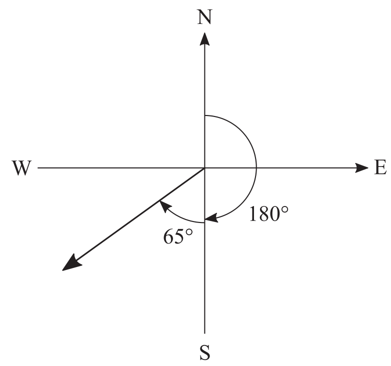
2. Measurement, STD2 M6 SM-Bank 3 MC



$$\begin{aligned} \text{True bearing} &= 180 + 65 \\ &= 245^{\circ} \end{aligned}$$

$\Rightarrow C$

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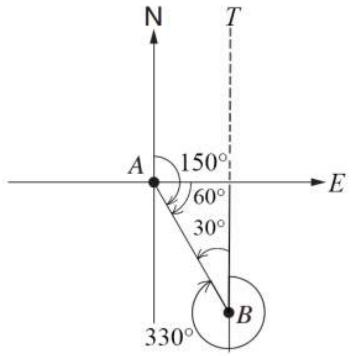
3. Trigonometry, 2ADV* T1 2010 HSC 10 MC



$\angle TBA = 30^\circ$ (angle sum of triangle)

$\therefore$ Bearing of A from B

$= 360 - 30$

$= 330^\circ$

$\Rightarrow D$

4. Trigonometry, 2ADV* T1 2012 HSC 20 MC

Using the cosine rule:

$$\begin{aligned}\cos \angle B &= \frac{a^2 + c^2 - b^2}{2ac} \\ &= \frac{59^2 + 80^2 - 31^2}{2 \times 59 \times 80} \\ &= 0.9449\ldots \\ \angle B &= 19^\circ \quad \text{(nearest degree)}\end{aligned}$$

$\therefore$ Bearing of Town C from B $= 180 - 19 = 161^\circ$

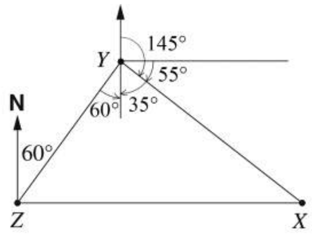
$\Rightarrow C$

5. Trigonometry, 2ADV* T1 2014 HSC 23 MC

COMMENT: Drawing a parallel North/South line through Y makes this question ***much simpler*** to solve.

Since X is due east of Z

$\Rightarrow$ Cannot be B or D



The diagram shows we can find

$\angle ZYX = 60 + 35^\circ = 95^\circ$

Using alternate angles (60°) and the 145° bearing of X from Y

$\Rightarrow C$

6. Trigonometry, 2ADV* T1 2016 HSC 25 MC

Using the sine rule,

$$\frac{\sin\angle BCA}{40} = \frac{\sin 25^\circ}{18}$$

$$\begin{aligned}\sin\angle BCA &= \frac{40 \times \sin 25^\circ}{18} \\ &= 0.939\ldots \\ \angle BCA &= 180 - 69.9 \quad (\angle BCA > 90^\circ) \\ &= 110.1^\circ\end{aligned}$$

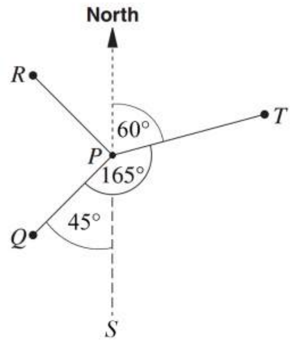
$\therefore$ Bearing of C from B

$= 110.1 + 25$ (external angle of triangle)

$= 135.1$

$\Rightarrow D$

7. Trigonometry, 2ADV* T1 2008 HSC 17 MC



$\angle QPS = 45^\circ \quad (Q \text{ is south west of } P)$

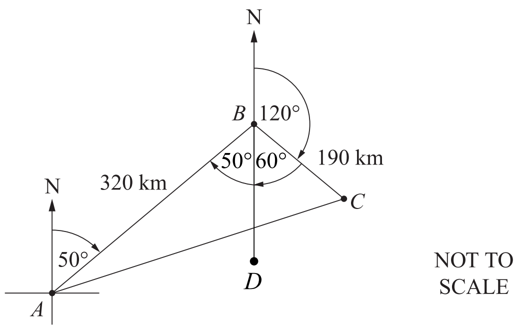
$\angle TPS = 165 – 45 = 120^\circ$

$\therefore\ \angle NPT = 60^\circ \quad (180^\circ \text{ in straight line})$

$\Rightarrow\ A$

8. Trigonometry, 2ADV T1 2018 HSC 12a

i.



Let D be south of B

$\angle ABD = 50^\circ \quad (\text{alternate angles})$

$\angle DBC = 60^\circ \quad (180^\circ \text{ in straight line})$

$\therefore \angle ABC = 50 + 60$

$= 110^\circ$

ii. Using the cosine rule:

$$\begin{aligned} AC^2 &= AB^2 + BC^2 – 2 \cdot AB \cdot BC \cdot \cos\angle ABC \\ &= 320^2 + 190^2 – 2 \times 320 \times 190 \times \cos 110^\circ \\ &= 180\ 089.64\ldots \\ \therefore AC &= 424.36\ldots \\ &= 420 \text{ km (nearest 10 km)} \end{aligned}$$

9. Trigonometry, 2ADV T1 2020 HSC 15

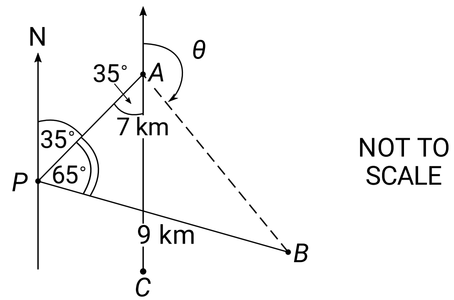
a. $\angle APB = 100 – 35$

$= 65^\circ$

b. Using cosine rule:

$$\begin{aligned} AB^2 &= AP^2 + PB^2 – 2 \times AP \times PB \cos 65^\circ \\ &= 49 + 81 – 2 \times 7 \times 9 \cos 65^\circ \\ &= 76.750\ldots \\ \therefore AB &= 8.760\ldots \\ &= 8.76 \text{ km (to 2 d.p.)} \end{aligned}$$

c.



$\angle PAC = 35^\circ \ (\text{alternate})$

Using cosine rule, find $\angle PAB$:

$$\cos\angle PAB = \frac{7^2 + 8.76 – 9^2}{2 \times 7 \times 8.76}$$

$= 0.3647\ldots$

$\therefore \angle PAB = 68.61\ldots^\circ$

$= 69^\circ \quad (\text{nearest degree})$

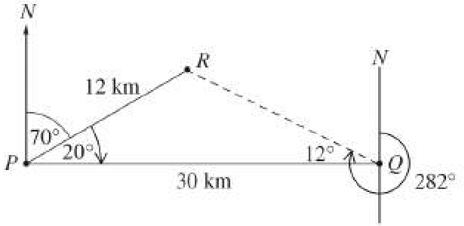
$\therefore$ Bearing of B from A (θ)

$= 180 – (69 – 35)$

$= 146^\circ$

10. Trigonometry, 2ADV T1 2004 HSC 3c

i. Join RQ to form ΔRPQ



$\angle RPQ = 90–70 = 20°$

Using the cosine rule:

$$\begin{aligned} RQ^2 &= PR^2 + PQ^2 – 2 \times PR \times PQ \times \cos\angle RPQ \\ &= 12^2 + 30^2 – 2 \times 12 \times 30 \times \cos 20° \\ &= 367.421\ldots \\ \therefore\ RQ &= 19.168\ldots \\ &= 19.2\ \text{km} \quad (1\ \text{d.p.}) \end{aligned}$$

ii. Using sine rule:

$$\begin{aligned} \frac{\sin\angle RQP}{12} &= \frac{\sin 20°}{19.168\ldots} \\ \sin\angle RQP &= \frac{12 \times \sin 20°}{19.168\ldots} \\ &= 0.214\ldots \\ \angle RQP &= 12.36\ldots° \\ &= 12° \quad \text{(nearest degree)} \end{aligned}$$

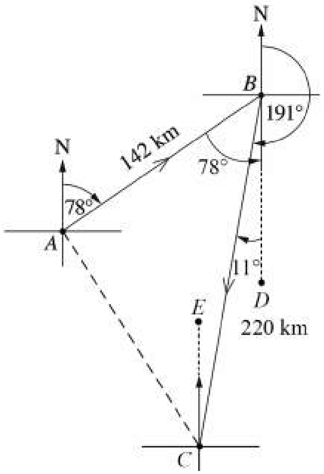
$\therefore$ Bearing of R from Q

$= 270 + 12$

$= 282°$

11. Trigonometry, 2ADV T1 2014 HSC 13d

i.



Find $\angle ABC$

Let D be south of B

$\therefore\ \angle CBD = 191 - 180 = 11^\circ$

$\angle DBA = 78^\circ$ (alternate)

$\angle ABC = 78 - 11$

$= 67^\circ$

Using Cosine rule:

$$AC^2 = AB^2 + BC^2 - 2 \cdot AB \cdot BC \cdot \cos\angle ABC$$
$$= 142^2 + 220^2 - 2 \times 142 \times 220 \times \cos 67^\circ$$
$$= 44\ 151.119\ldots$$
$$\therefore\ AC = 210.121\ldots$$
$$\approx 210 \text{ km} \quad \ldots \text{as required}$$

ii. Find $\angle ACB$

Using Sine rule:

$$\frac{\sin\angle ACB}{142} = \frac{\sin\angle ABC}{210}$$
$$\sin\angle ACB = \frac{142 \times \sin 67^\circ}{210}$$
$$= 0.6224\ldots$$
$$\angle ACB = 38.494\ldots$$
$$= 38^\circ \text{ (nearest degree)}$$

Let E be due North of C

$\angle ECB = 11^\circ$ (alternate to $\angle CBD$)

$\therefore\ \angle ECA = 38 - 11$

$= 27^\circ$

$\therefore$ Bearing of A from C

$= 360 - 27$

$= 333^\circ$

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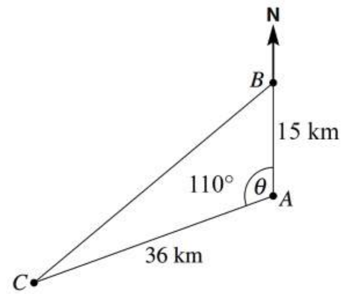
12. Trigonometry, 2ADV* T1 2005 HSC 27c

i. There is 360° about point A

$$\therefore\ \theta + 250^\circ = 360^\circ$$
$$\theta = 110^\circ$$

ii.



$$a^2 = b^2 + c^2 - 2ab\cos A$$
$$CB^2 = 36^2 + 15^2 - 2 \times 36 \times 15 \times \cos 110^\circ$$
$$= 1296 + 225 - (-369.38\ldots)$$
$$= 1890.38\ldots$$
$$\therefore CB = 43.47\ldots$$
$$= 43 \text{ km (nearest km)}$$

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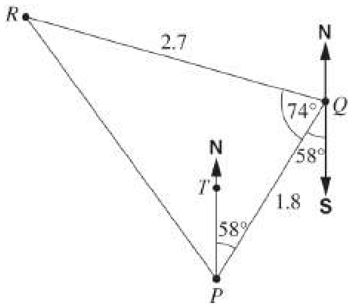
13. Trigonometry, 2ADV* T1 2009 HSC 27b

i.



$\angle PQS = 58^\circ \quad$ (alternate to $\angle TPQ$)

Bearing of R from Q

$= 180^\circ + 58^\circ + 74^\circ$

$= 312^\circ$

TIP: Draw North-South parallel lines through relevant points to help calculate angles as shown in the Worked Solutions.

(ii) Using Cosine rule:

$$RP^2 = RQ^2 + PQ^2 - 2 \times RQ \times PQ \times \cos \angle RQP$$

$$= 2.7^2 + 1.8^2 - 2 \times 2.7 \times 1.8 \times \cos 74^\circ$$

$$= 7.29 + 3.24 - 2.679\ldots$$

$$= 7.851\ldots$$

$$\therefore RP = \sqrt{7.851\ldots}$$

$$= 2.8019\ldots$$

$$\approx 2.8 \text{ km } (1 \text{ d.p.})$$

(iii) Using $A = \dfrac{1}{2} ab\sin C$

$$A = \frac{1}{2} \times 2.7 \times 1.8 \times \sin 74^\circ$$

$$= 2.3358\ldots$$

$$= 2.3 \text{ km}^2$$

$\therefore$ No-go zone is 2.3 km²

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14. Trigonometry, 2ADV* T1 2007 26a

i. 2 hrs 48 mins $= 168$ mins

$$\text{Speed } (A \text{ to } Q) = \frac{15}{168}$$

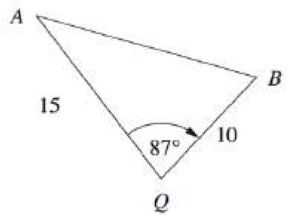
$$= 0.0892\ldots \text{ km/min}$$

$$\text{Speed (in km/hr)} = 0.0892\ldots \times 60$$

$$= 5.357\ldots \text{ km/hr}$$

$$= 5 \text{ km/hr (nearest km/hr)}$$

ii.



Using cosine rule

$$AB^2 = 15^2 + 10^2 - 2 \times 15 \times 10 \times \cos 87^\circ$$

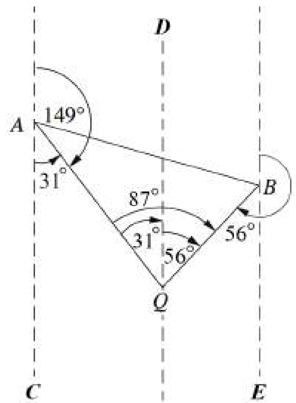
$$= 309.299\ldots$$

$$AB = 17.586\ldots$$

$$= 18 \text{ km (nearest km)}$$

$\therefore$ The distance from Town A to Town B is 18 km.

iii.



$\angle CAQ = 31^\circ \quad$ (straight angle at A)

$\angle AQD = 31^\circ \quad$ (alternate angle $AC \parallel DQ$)

$\angle DQB = 87 - 31 = 56^\circ$

$\angle QBE = 56°$ (alternate angle $DQ \parallel BE$)

$\therefore$ Bearing of Q from B

$= 180 + 56$

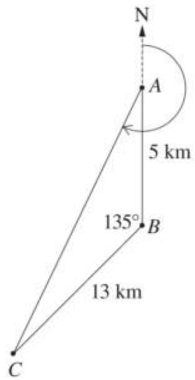
$= 236°$

15. Trigonometry, 2ADV* T1 2017 HSC 30c

(i) Using cosine rule:

$$AC^2 = AB^2 + BC^2 - 2 \times AB \times BC \times \cos 135°$$
$$= 5^2 + 13^2 - 2 \times 5 \times 13 \times \cos 135°$$
$$= 285.923\ldots$$
$$\therefore AC = 16.909\ldots$$
$$= 17 \text{ km (nearest km)}$$

(ii)



Using sine rule, find $\angle BAC$:

$$\frac{\sin\angle BAC}{13} = \frac{\sin 135°}{17}$$
$$\sin\angle BAC = \frac{13 \times \sin 135°}{17}$$
$$= 0.5407\ldots$$
$$\angle BAC = 32.7°$$

♦♦ Mean mark part (ii) 31%.

$\therefore$ Bearing of C from A

$= 180 + 32.7$

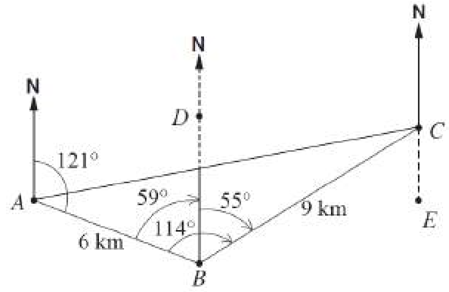
$= 212.7°$

$= 213°$

16. Trigonometry, 2ADV* T1 2011 HSC 24c

STRATEGY: This deserves repeating again: *Draw North-South parallel lines through major points to make the angle calculations easier!*

i.



Let point D be due North of point B

$$\angle ABD = 180 - 121 \text{ (cointerior with } \angle A)$$
$$= 59°$$
$$\angle DBC = 114 - 59$$
$$= 55°$$

$\therefore$ Bearing of C from B is $055°$

ii. Using cosine rule:

$$AC^2 = AB^2 + BC^2 - 2 \times AB \times BC \times \cos\angle ABC$$
$$= 6^2 + 9^2 - 2 \times 6 \times 9 \times \cos 114°$$
$$= 160.9275\ldots$$
$$\therefore AC = 12.685\ldots \quad (\text{Noting } AC > 0)$$
$$= 13 \text{ km (nearest km)}$$

iii. Need to find $\angle ACB$ (see diagram)

$$\cos\angle ACB = \frac{AC^2 + BC^2 - AB^2}{2 \times AC \times BC}$$
$$= \frac{(12.685\ldots)^2 + 9^2 - 6^2}{2 \times (12.685..) \times 9}$$
$$= 0.9018\ldots$$
$$\angle ACB = 25.6° \text{ (to 1 d.p.)}$$

MARKER'S COMMENT: The best responses showed clear working on the diagram.

From diagram,

$\angle BCE = 55°$ (alternate angle, $DB \parallel CE$)

$\therefore$ Bearing of A from C

$= 180 + 55 + 25.6$

$= 260.6$

$= 261°$ (nearest degree)